

Paul Lee

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EDUCATION

The University of Texas at Austin | Austin, TX

Expected May 2028

Bachelor of Science, Mechanical Engineering | GPA 3.82

- Certificate in Programming and Computation
- Relevant Coursework: Thermodynamics, Dynamics, Statics, Engineering Computing Methods, Software Design, Elements of Database, and Electricity and Magnetism

ENGINEERING EXPERIENCE

Celestica *Mechanical Engineering Intern* | Dallas, TX

May 2026 - August 2026

- Redesigned a sheet metal fan tray with a custom sliding mechanism, saving ~3.5 metric tons of steel (~\$10K in raw material cost) across a 60K-unit production run
- Designed sheet metal chassis, PCBA component placements, and power cable routing in Creo, reducing assembly time by up to 50% across AI server assemblies
- Revised 150+ part and assembly drawings, applying GD&T and ballooning inspection-critical dimensions
- Supported validation of blades through thermal, leak, seismic, tilt, and roll testing to verify product performance
- Performed worst-case and RSS tolerance stackup analysis and global interference checks across large TLA assemblies

MINIMAX Lab *Undergraduate Research Assistant* | Austin, TX

Sept 2025 - Feb 2026

- Modeled a 125 μm silicon AFM cantilever in COMSOL, extracting stress distribution, spring constant, and the first three flexural modes for probe calibration
- Reviewed literature on active AFM cantilevers and piezoelectric actuation to inform model parameters

DUCK Lab *Research Assistant* | Austin, TX

Oct 2024 - May 2025

- Increased annual energy yield by 8.2% and projected profit by 40.7% through solar farm layout optimization
- Developed and benchmarked 50+ solar farm layouts in Aladdin, cutting design generation time by 11x
- Conducted controlled studies on tree count, building placement, and polygon geometry, producing design guidelines adopted by the lab

PROJECTS

Hero Robot

- Redesigned the chassis belly pan and box tube, redimensioning all misaligned mounting holes across the assembly
- Designed and prototyped rounded skirt geometries that reduced collision forces by ~40%

Crates and 1V1 Arena

- Designed crates and 1v1 arena models, integrating McMaster-Carr components for precise assembly
- Fabricated wooden barriers for competition using miter saws, planers, and drills
- Built a 50 ft x 50 ft PVC testing arena from CAD designs, reducing robot testing setup time by ~30%

Robotathon

- Developed an autonomous robot capable of color detection, line following, and obstacle navigation.
- Designed and fabricated custom chassis components using 3D printing, including battery and IR sensor mounts.

TECHNICAL SKILLS

- **Software:** Solidworks, Creo, ANSYS, COMSOL, Python, PLM, EZ-tol, Aladdin, Excel
- **Technical:** CAD, FEA, 2D Drawing, 3D Printing, DFMA, Tolerance Analysis, CFD, Woodworking, GD&T, Soldering
- **Languages:** English (Native), Korean (Native)
- **Certificates:** CSWA, CornellX(FEA/CFD), CAM and Design Manufacturing(Fusion 360)

HONORS & AWARDS

- University Honors (Fall 2024, Spring 2025)
- 1st place, Robotathon 2024
- BC Achievement Scholarship Award